Name of the Student: _____ **Roll no:** _____

CLASS: - T.E (CSE) **Division:** **Course: - PCC305CSE- Machine Learning Lab**

Experiment No. 10

**** Ensemble Learning: Random Forest And Boosting****

Marks:

Date of Performance: / /2026 **Sign with Date:**

Aim: To analyze the performance of ensemble learning techniques such as Random Forest and Boosting on a suitable open-access dataset in comparison with a baseline machine learning model.

Lab Objective: Enable students to load and preprocess a real-world dataset, train a baseline classifier and ensemble classifiers, evaluate their performance using suitable classification metrics, compare the results, and interpret the benefits and limitations of ensemble learning..

Dataset: Breast Cancer Wisconsin (Diagnostic) dataset, available through scikit-learn (sklearn.datasets.load_breast_cancer). It contains 569 samples, 30 numerical features computed from digitized images of breast mass fine-needle aspirates, and two target classes: malignant and benign.

Suggested source: Built into scikit-learn through load_breast_cancer().

Why this dataset?

It is open-access, compact, numeric, and suitable for binary classification. A stratified train-test split supports fair comparison of the models.

Tools / Libraries Required

- Python 3.x
- Jupyter Notebook / Google Colab
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

Theory

Ensemble learning combines predictions from multiple models to improve predictive performance, stability, or generalization compared with relying on a single model. This experiment compares Logistic Regression as a baseline with Random Forest and Gradient Boosting classifiers.

1. Baseline Model – Logistic Regression

Logistic Regression is a supervised classification algorithm that estimates class probabilities using the logistic function. It provides a useful reference model because it is relatively simple and interpretable. Features are standardized before fitting.

2. Random Forest

Random Forest is a bagging-based ensemble that trains multiple decision trees on bootstrap samples and considers random subsets of features when splitting. For classification, the trees vote or average class probabilities. This can reduce variance and help capture nonlinear relationships.

3. Gradient Boosting

Gradient Boosting builds a sequence of weak learners, commonly shallow decision trees. Each new learner is fitted to improve errors made by the current ensemble, optimizing a loss function through gradient-based updates. It can achieve strong predictive performance but may require careful tuning.

4. Train-Test Split and Evaluation

The dataset is divided into training and testing subsets using an 80:20 split with stratification and a fixed random state. The models are trained only on the training data and evaluated on the held-out test data.

Metric	Formula / Meaning	Purpose
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Overall fraction of correct predictions.
Precision	TP / (TP + FP)	Reliability of positive predictions.
Recall	TP / (TP + FN)	Ability to identify actual positive cases.
F1-score	2 × (Precision × Recall) / (Precision + Recall)	Harmonic mean of precision and recall.
Training time	Elapsed time to fit the model	Compares computational cost.

For this dataset, malignant is treated as the positive class (label 0 in scikit-learn). Report precision, recall, and F1 using pos_label=0, or report per-class metrics to avoid ambiguity.

Algorithm

Input: Breast Cancer Wisconsin dataset with 569 observations and 30 numerical features.
Output: Test-set evaluation metrics and a comparison of Logistic Regression, Random Forest, and Gradient Boosting.

1. Start.
2. Import required Python libraries and evaluation functions.
3. Load the dataset using load_breast_cancer().
4. Separate the feature matrix X and target vector y.
5. Split the data into 80% training and 20% testing subsets using stratify=y and random_state=42.
6. Create a pipeline containing StandardScaler and LogisticRegression for the baseline model.
7. Initialize RandomForestClassifier and GradientBoostingClassifier with fixed random_state values.
8. Train each model on the training data and record training time.
9. Predict labels for the test data.
10. Calculate accuracy, malignant-class precision, recall, F1-score, and training time for each model.
11. Display the comparison in a table and plot the metrics.
12. Interpret which model performs best and discuss the trade-off between performance and training time.
13. Stop.

Expected Observation

Random Forest and Gradient Boosting may outperform the Logistic Regression baseline on one or more metrics because they can model nonlinear patterns. Actual values depend on the split, library version, and model settings; students must report the values produced by their execution. In a medical screening context, malignant-class recall is especially important, and accuracy alone should not determine the preferred model.

Sample Results / Observation

Model	Accuracy	Precision (Malignant)	Recall (Malignant)	F1 (Malignant)	Training Time (s)
Logistic Regression (Baseline)					
Random Forest					
Gradient Boosting					

Result

The performance of Logistic Regression, Random Forest, and Gradient Boosting was evaluated on the Breast Cancer Wisconsin dataset using accuracy, malignant-class precision, recall, F1-score, and training time. The best-performing model based on the selected evaluation priority was _____.

Questions:

1. What is ensemble learning, and why is it used?
2. Differentiate between bagging and boosting.
3. Explain the working principle of Gradient Boosting.
4. Why is stratified sampling used during train-test splitting?
5. Why should precision, recall, and F1-score be considered in addition to accuracy?

Attendance and Timely Submission (10)	**Execution (10)**	**Observations and Records (10)**	**Journal Documentation (10)**	**Oral (10)**